No. of Printed Pages : 4 **MCS–011**

MASTER OF COMPUTER APPLICATIONS (Revised) (MCA)

Term-End Examination

June, 2024

MCS–011 : PROBLEM SOLVING AND PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *(i) Question no. **1** is compulsory.*

*(ii) Answer any **three** questions from the rest.*

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1. (a) Write an algorithm and draw corresponding flowchart to print first n even natural numbers and find their sum. 10

(b) Define recursion. Explain it with *one* example of C code segment. 6

(c) Write a program to generate the following pattern : 8

1			
2	3		
4	5	6	
7	8	9	10

(d) Find the square and cube of any given number using Macro. 5

(e) Define a structure. How do we declare and use the structure ? Explain with an example. 5

(f) Explain the following looping constructs in C by giving code segment of each : 6

(i) While

(ii) For

2. (a) Explain a conditional operator in C. Write a program in C to find the greater number between two numbers using conditional operator. 6

(b) Write a C program to calculate income tax of an employee based on annual salary : 8

If salary < 5 lakhs	:	No tax
5 lakhs – 8 lakhs	:	10% tax
8 lakhs – 10 lakhs	:	25% tax
Salary > 10 lakhs	:	30% tax

(c) Define a function in C. List and explain various categories of functions. 6

3. (a) Simulate the calculator to perform arithmetic operations add, subtract, multiplication, division on two variables depending on the choice selected by the user. 10

(b) Write a program to find maximum marks among the marks of 10 students using arrays. 10

4. (a) Describe the initialization, declaration of pointer, accessing a variable using pointer with the help of an example. 5

(b) Differentiate between sequential and random file access. Also write inbuilt functions in C related to file access. 10

(c) Write a C program to obtain GCD of two numbers. 5

5. Write short notes on the following : $5 \times 4 = 20$

(i) Storage classes in C

(ii) Preprocessor directives

(iii) Precedence of operators

(iv) Typecasting

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MASTER OF COMPUTER APPLICATIONS/BACHELOR OF COMPUTER APPLICATIONS (REVISED) (MCA/BCA)

Term-End Examination

June, 2023

MCS-011 : PROBLEM SOLVING AND PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question No. **1** is compulsory. Answer any **three** questions from the rest.*

1. (a) Write an algorithm to calculate simple interest. Also draw flow chart for this algorithm (**Hint :** SI = (PTR)/100). 10

   (b) Write a C program to create a structure to store name, rollnumber, address and course (BCA, MCA) of ten students. Use array of structure to display details of the students.

10

(c) Write a C program which read an array of ten integer values and search a given value in the array. If that value exist in the array display its square otherwise display. "The value is missing." 10

(d) Write a program in 'C' using printers to find the length of a given string. 10

2. (a) Write a C program to find the sum of the following series upto 20 terms : 6

$$1 + 4 + 7 + 10 + 13 \text{ ....}$$

(b) Briefly explain *two* categories of constants in C. 4

(c) Write a C program for multiplication of two 4×4 matrices. 10

3. (a) Explain the difference between a++ and ++a. Also find the value that would be printed by the following code : 6

```
    int a = 4;
    int b = 2;
    int c = 0;
a = c++;
a = a + b++;
printf ("a = %d", a);
```

(b) Explain the use of conditional operator with the help of an example. 4

(c) What is for loop ? How is it different from do .... while loop ? Explain with the help of an example. Also explain use of “break statement”. 10

4. (a) Write a ‘C’ program, using structures, to calculate the gross salary and net salary if Basic, D. A., T. A. and deduction are given as input. 10

(b) Write the syntax and explain the use of the following functions in C : $2\times4=8$

(i) return

(ii) malloc

(iii) isupper

(iv) fclose

(c) Define a macro to find the cube of a given number. 2

5. (a) Write C program which take a string as input and print it in reverse order, without using any in-built function. 5

(b) Write a C program to check whether a file whose name is given exist or not in the given list of file names. 5

(c) Explain the following with the help of a suitable example for each : $4\times2\frac{1}{2}=10$

(i) File access modes

(ii) Array of pointers

(iii) Automatic variable

(iv) Size of operator

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M. C. A. (REVISED)/B. C. A. (REVISED)

(MCA/BCA)

Term-End Examination

June, 2020

MCS-011 : PBOBLEM SOLVING AND PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *(i) Question No. 1 is compulsory.*

(ii) Answer any ***three*** *questions from the rest.*

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1. (a) Give the equivalent C expression for the following algebraic expression : $2 \times 3 = 6$

(i) $\dfrac{a^2b^3c^4 - d^4}{e(m-n)}$

(ii) $xy - \left[(p+q)^4 / r^2\right]$

(b) Write a recursive function in C to generate a Fibonacci series. 6

(c) Write an algorithm and draw a corresponding flow chart to check whether the given number is prime or not. 8

(d) Write an interactive C program to perform the following operation on a 3 × 3 matrix with appropriate validation checks : 10

$$C = A * B$$

(e) Explain the differences between static, auto, register and global variables with an example for each. $2\frac{1}{2} \times 4 = 10$

2. (a) List the arithmetic, logical and relational operators in C. When a, b, c and d are integers and values of a, b and c are 8, 6 and 4 respectively. Find the value of d, 8

if $d = a + (b - c) * a / c$

(b) Write a program in C to generate the following pattern : 6

```
1
1 2
1 2 3
1 2 3 4
```

(c) Differentiate between call-by value and call-by reference using an example program for each. 6

3. (a) Using pointers, write a program in C to count the no. of occurrence of each character in a given string. 10

Sample I/P : ARRANGE

O/P :

A – 2 times G – 1 time

R – 2 times E – 1 time

N – 1 time

(b) Write a C program to calculate the income tax for 5 employees if name of the employee, designation, department and annual salary is given as per the following slab-rate : 10

Below 5 lakhs	:	No tax
Above 5 lakhs and below 7 lakhs	:	10% tax on annual income
Above 7 lakhs and below 10 lakhs	:	20% tax on annual income
Above 10 lakhs	:	30% tax on annual income

4. (a) Using file handling concept, write a C program to read a file and count the no. of lines in the file. 10

(b) What are Unions ? Give an example code segment to initialize a union and to access a member of a union. Mention the difference between a structure and a union. 10

5. (a) Explain the following conditional statements with an example for each : 10

(i) If .... else statement

(ii) If statement

(iii) Nested if .... else statement

(iv) Else .... if ladder

(b) Define a function in C. List and explain various categories of functions. Also, illustrate a function to find square root of a given number. 10

MCA (Revised) / BCA (Revised)
Term-End Examination
June, 2022

MCS-011 : PROBLEM SOLVING AND PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage : 75%)

Note : *Question no.* ***1*** *is* ***compulsory****. Answer any* ***three*** *questions from the rest.*

1. (a) Draw a flowchart and write an algorithm to calculate the factorial of a given number. *10*

(b) Write a program to display string 'INFINITY' in the following pattern (using "LOOP" control statement) : *10*

I
I N
I N F
I N F I
I N F I N
I N F I N I
I N F I N I T
I N F I N I T Y

(c) Write a program to generate Fibonacci series using recursion. *10*

(d) Write an interactive C program for each to illustrate the following concepts : *10*

(i) Enumerated data type

(ii) Macros in C

(iii) Typedef

(iv) Goto statement

2. (a) Write an algorithm and draw the corresponding flowchart to calculate whether the given number is prime or not. *10*

(b) Write a C program to perform the following operation on matrices :

$$D = (A*B) + C$$

where A, B and C are matrices of 3×3 size and D is the resultant matrix. *10*

3. (a) Write a program to find the minimum marks among the given marks of 20 students. *10*

(b) Write a program to find the string length without using strlen( ) function. *10*

4. (a) Define function. How are functions declared in C language ? What are function prototypes and what is the use of **return** statements ? *10*

(b) Explain the function "call by reference" in C language. Give advantages and disadvantages of it. *10*

5. (a) Explain the following statements with the help of an example for each : *10*

(i) Break

(ii) Goto

(iii) Continue

(b) Using File Handling Concept, write a program to copy one file to another. *5*

(c) Write a short note on Ternary Operator with an illustration. *5*

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MCS-011

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2021

MCS-011 : PROBLEM SOLVING AND PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage : 75%)

Note : *Question no.* ***1*** *is* ***compulsory****. Answer any* ***three*** *questions from the rest.*

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1. (a) Write an algorithm and draw a corresponding flow chart to check whether the given year is a leap year or not. *10*

(b) Write a C function **isodd(num**) that accepts an integer argument and returns 1 if the argument is odd and 0 otherwise. *4*

(c) Write a C program that invokes this function to generate numbers between the given range. *6*

(d) Define an Array. How do we declare and initialize a single-dimensional array and a 2-dimensional array ? Explain with an example for each. *4*

(e) Write a C program to implement STRING COPY operation that copies a string "**str1**" to another string "**str2**" without using library function. *8*

(f) Write a C program to mantain a record of "n" student details using an array of structures with structure-variables (Rollnumber, Name, Marks1, Marks2, Marks3, Marks4 and Grade). Define each field of the structure with appropriate datatype. Print the marks of the student of given Rollnumber as input. *8*

2. (a) Discuss dynamic memory allocation in C. Write and explain dynamic allocation functions in C. *6*

(b) Write and explain any two pre-processor directives in C. *4*

(c) Write a C program to find the sum of diagonal elements of a 3×3 matrix. *10*

3. (a) Write a C program to create a file of numbers given by the user and copy odd numbers to odd.dat file and even numbers to even.dat file. *10*

(b) *Using recursion*, write a C program to find the factorial of a given number. *10*

4. (a) Write a function to print the sum of the following series :

$$1 + 2^2 + 3^3 + ... + n^n$$

where "n" is passed as an argument to the function. *8*

(b) Write a program to calculate the number of vowels (a, e, i, o, u) separately in the entered string and display their individual count. *12*

5. (a) Write two differences between a structure and a union with appropriate examples. *4*

(b) Write a C program to find the substring in a string without using a library function. *10*

(c) Write a program to delete an element from a given location of an array of integers. *6*